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SOE-BVAR Suite — Model Scorecard

Data source: real (RBA + ABS + FRED)

Panel: 1997Q4-2026Q1 | **Targets:** gdp_growth, cpi_inflation, unemp_rate, cash_rate | **Horizons:** 1-12 quarters

Evaluation: expanding-window pseudo-real-time, 36 forecast origins; densities scored by CRPS and log predictive density, points by RMSE, all by horizon.

Two views, level first. Performance is reported two ways. The headline is the **level** error (§2): the forecast error of each series' *level* at quarter $t+h$ — for GDP and inflation (modelled as growth) the cumulative level from the forecast origin, where real GDP and the price level land, and for unemployment and the cash rate the rate level itself. Alongside it (§3) is the **quarterly-growth** view: each target's single-quarter outcome at $t+h$. The year-ended *growth* scores remain in [output/tables/scores_by_horizon.csv](#).

Diagnostics (§9): all green (block exogeneity, MCMC convergence, forecast sanity, no-look-ahead, reproducibility).

1. The models

Every VAR member is **block-exogenous** (the domestic block never feeds back into the foreign block) and produces **iterated** density forecasts. Members are designed to fail differently; see README.md for the full rationale.

1a. VAR members

Model	Family	System	Lags	Shrinkage λ	Volatility	COVID
<code>small_min</code>	Independent Normal-inverse- Wishart (Gibbs) + SOC/DIO	small (8 var)	4	auto (GLP)	constant	LP scaling
<code>small_ss</code>	Steady-state (Vil- lani)	small (8 var)	4	auto (GLP)	constant	LP scaling
<code>small_sv</code>	Stochastic volatility, equa- tion-by-equation	small (8 var)	4	auto (GLP)	stochastic (SV)	t-errors (SV-t)
<code>small_loose_p5</code>	Independent Normal-inverse- Wishart (Gibbs)	small (8 var)	5	0.4	constant	LP scaling
<code>medium_min</code>	Stochastic volatility, equa- tion-by-equation	medium (13 var)	2	auto (GLP)	stochastic (SV)	t-errors (SV-t)
<code>medium_conj</code>	Block-recursive conjugate NIW + SOC/DIO	medium (13 var)	4	0.1	constant	LP scaling
<code>small_tight</code>	Block-recursive conjugate NIW + SOC/DIO	small (8 var)	4	0.05	constant	LP scaling

1b. Benchmark members (the bar every VAR must clear)

Model	Description	COVID
<code>rw</code>	Random walk	LP scaling
<code>ar4</code>	Bayesian AR(4)	LP scaling

Model	Description	COVID
ucsv	Unobserved components + stochastic volatility	t-errors (robust)
ucmean	Unconditional mean	LP scaling

1c. Combination schemes (density pools)

Weights estimated **per target variable and per horizon bucket** (near 1-4, medium 5-8, far 9-12), shrunk toward equal weights, strictly recursive (no look-ahead).

Scheme	How weights are set
combo_equal	Equal weights (the benchmark pool — hard to beat)
combo_logscore	Recursive log-score weights, with forgetting
combo_pool	Optimal prediction pool (Hall-Mitchell / Geweke-Amisano)
combo_bma	Bayesian model averaging — reported as a diagnostic only

2. Forecast performance — levels (the headline)

These tables score the **level** of each series at t+h: the cumulative real GDP and price level for the growth-modelled variables, the rate level for unemployment and the cash rate. Level errors accumulate the whole forecast path, so they grow with horizon and are not comparable across horizons — read down each column, not across. Lower CRPS / RMSE is better; higher log score is better. **Bold** = best in that column; models ordered best-first (mean over the shown horizons).

Best single model by variable and horizon bucket (CRPS) — lowest mean CRPS in the bucket; value in parentheses:

Variable	near (1-4)	medium (5-8)	far (9-12)
gdp_growth	medium_minn (0.985)	small_sv (1.665)	small_sv (2.102)
cpi_inflation	rw (0.386)	rw (1.358)	ar4 (2.693)
unemp_rate	medium_minn (0.325)	small_ss (0.631)	small_ss (0.726)
cash_rate	small_sv (0.307)	combo_equal (0.894)	ar4 (1.447)

Density accuracy by variable and horizon (CRPS, lower better)

Real GDP growth (qtr %) ([gdp_growth](#)) — cumulative level from the origin

Model	h=1	h=4	h=8	h=12
small_sv	0.604	1.269	1.921	2.154
medium_minn	0.614	1.266	1.977	2.315
combo_equal	0.631	1.300	2.042	2.233
small_tight	0.682	1.366	2.101	2.165
ucsv	0.629	1.298	2.026	2.391
medium_conj	0.672	1.383	2.139	2.238
small_loose_p5	0.628	1.362	2.161	2.300
small_ss	0.662	1.376	2.190	2.284
small_minn	0.677	1.414	2.230	2.293
ar4	0.668	1.342	2.156	2.470
ucmean	0.680	1.380	2.273	2.662
combo_pool	0.635	1.432	2.438	3.699
combo_logscore	0.760	1.803	4.328	3.982

Model	h=1	h=4	h=8	h=12
combo_bma	0.840	2.232	6.320	5.346
rw	0.941	3.264	6.711	11.084

Trimmed-mean CPI inflation (qtr %) ([cpi_inflation](#)) – cumulative level from the origin

Model	h=1	h=4	h=8	h=12
rw	0.131	0.674	1.832	3.356
ar4	0.162	0.844	1.973	3.108
combo_equal	0.160	0.770	1.912	3.323
combo_pool	0.147	0.764	2.081	3.502
combo_logscore	0.139	0.745	2.037	3.614
small_minn	0.170	0.824	2.109	3.714
small_ss	0.172	0.849	2.165	3.655
small_sv	0.171	0.856	2.153	3.769
ucmean	0.230	1.056	2.269	3.405
medium_minn	0.165	0.810	2.153	4.021
small_tight	0.198	0.925	2.241	3.910
combo_bma	0.146	0.808	2.391	3.945
medium_conj	0.191	0.895	2.234	4.029
ucsv	0.176	0.926	2.333	4.058
small_loose_p5	0.193	0.944	2.437	4.089

Unemployment rate (%) ([unemp_rate](#)) – rate level at t+h

Model	h=1	h=4	h=8	h=12
small_ss	0.148	0.497	0.690	0.747
combo_equal	0.159	0.493	0.702	0.785
medium_minn	0.150	0.477	0.726	0.803
small_sv	0.157	0.501	0.743	0.771
small_minn	0.150	0.524	0.744	0.789
small_loose_p5	0.150	0.540	0.775	0.812
small_tight	0.158	0.549	0.779	0.811
combo_pool	0.177	0.496	0.767	0.860
ucsv	0.272	0.536	0.740	0.841
rw	0.169	0.545	0.777	0.913
medium_conj	0.163	0.545	0.820	0.896
ar4	0.189	0.534	0.760	0.955
combo_logscore	0.383	0.618	0.776	0.841
combo_bma	0.677	0.806	0.938	0.915
ucmean	0.726	0.774	0.905	1.034

Cash rate (%) ([cash_rate](#)) – rate level at t+h

Model	h=1	h=4	h=8	h=12
combo_equal	0.148	0.514	1.120	1.652
small_ss	0.140	0.526	1.149	1.673
ar4	0.167	0.589	1.141	1.593
rw	0.183	0.590	1.136	1.670
small_minn	0.144	0.528	1.152	1.792
combo_pool	0.129	0.549	1.225	1.715
combo_logscore	0.125	0.538	1.241	1.780
combo_bma	0.128	0.576	1.381	1.854
ucsv	0.175	0.646	1.281	1.842
small_tight	0.157	0.598	1.286	1.935
small_loose_p5	0.162	0.593	1.247	2.038
medium_conj	0.192	0.572	1.239	2.053
small_sv	0.113	0.535	1.410	2.147
medium_minn	0.112	0.559	1.465	2.494
ucmean	1.563	1.654	1.671	1.615

Point accuracy by variable and horizon (RMSE, lower better)

gdp_growth

Model	h=1	h=4	h=8	h=12
small_sv	1.573	2.524	3.599	3.938
small_tight	1.614	2.623	3.654	3.830
small_ss	1.603	2.587	3.693	3.878
small_minn	1.619	2.633	3.715	3.910
small_loose_p5	1.571	2.585	3.674	4.071
ar4	1.632	2.632	3.631	4.036
ucsv	1.618	2.587	3.689	4.106
ucmean	1.619	2.634	3.685	4.152
medium_conj	1.616	2.690	3.774	4.030
medium_minn	1.605	2.625	3.805	4.323
combo_equal	1.624	2.815	4.140	4.971
combo_pool	1.688	3.411	5.770	8.846
combo_logscore	2.059	5.597	11.640	8.842
combo_bma	2.308	6.971	15.317	11.148
rw	2.377	7.807	15.379	24.092

cpi_inflation

Model	h=1	h=4	h=8	h=12
ar4	0.315	1.433	2.905	4.138
ucmean	0.434	1.648	3.084	4.269

Model	h=1	h=4	h=8	h=12
combo_equal	0.327	1.500	3.319	5.208
combo_pool	0.287	1.450	3.451	5.449
combo_logscore	0.269	1.410	3.420	5.675
small_ss	0.333	1.587	3.541	5.413
small_minn	0.333	1.590	3.549	5.525
rw	0.235	1.301	3.486	6.042
small_tight	0.402	1.736	3.655	5.611
small_sv	0.360	1.671	3.716	5.907
medium_conj	0.400	1.750	3.777	5.951
medium_minn	0.340	1.591	3.729	6.223
ucsv	0.361	1.697	3.812	6.017
combo_bma	0.278	1.516	3.914	6.180
small_loose_p5	0.371	1.735	3.926	6.181

unemp_rate

Model	h=1	h=4	h=8	h=12
small_ss	0.339	0.890	1.208	1.226
ar4	0.413	0.915	1.174	1.300
combo_equal	0.370	0.903	1.259	1.324
small_sv	0.354	0.927	1.333	1.349
small_minn	0.348	0.946	1.337	1.364
combo_pool	0.428	0.919	1.260	1.401
small_tight	0.362	0.953	1.351	1.371
small_loose_p5	0.349	0.989	1.382	1.467
medium_minn	0.341	0.943	1.388	1.516
ucsv	0.614	0.991	1.281	1.342
rw	0.391	0.984	1.391	1.509
medium_conj	0.375	1.050	1.487	1.574
combo_logscore	0.880	1.126	1.304	1.390
ucmean	1.260	1.326	1.435	1.542
combo_bma	1.252	1.365	1.502	1.448

cash_rate

Model	h=1	h=4	h=8	h=12
small_ss	0.276	0.990	1.908	2.581
ar4	0.307	1.103	1.931	2.467
combo_equal	0.341	0.971	1.916	2.777
combo_logscore	0.272	0.990	2.133	2.862
small_minn	0.292	1.051	2.045	2.902

Model	h=1	h=4	h=8	h=12
rw	0.352	1.194	2.062	2.684
combo_pool	0.290	1.025	2.149	2.849
ucsv	0.358	1.208	2.080	2.695
combo_bma	0.282	1.054	2.353	2.961
small_tight	0.313	1.190	2.230	3.078
small_sv	0.252	0.982	2.306	3.468
small_loose_p5	0.352	1.113	2.180	3.387
medium_conj	0.479	1.218	2.280	3.418
medium_minn	0.265	1.052	2.486	4.079
ucmean	2.576	2.658	2.649	2.635

Density calibration — mean log predictive density (higher better)

Averaged across the 4 targets. $-\infty$ marks at least one origin where the realization fell outside a member's predictive support; the combinations, which always assign positive density, never do.

Model	h=1	h=4	h=8	h=12
combo_equal	-1.844	-1.570	-2.328	-2.764
combo_pool	-1.816	-1.674	-2.502	-2.933
combo_logscore	-1.925	-1.698	-2.571	-2.941
combo_bma	-3.731	-3.547	-3.463	-3.201
small_sv	-4.619	-3.748	-4.950	-3.554
small_loose_p5	-6.452	-6.500	-10.199	-6.380
ar4	$-\infty$	-4.390	-5.271	-6.282
medium_conj	$-\infty$	-4.830	-10.309	-10.904
medium_minn	$-\infty$	-3.713	-3.342	-3.985
rw	$-\infty$	-4.619	-4.431	-4.180
small_minn	$-\infty$	-5.758	-11.007	-10.271
small_ss	$-\infty$	-5.976	-10.826	-8.541
small_tight	$-\infty$	-9.674	-15.438	-14.541
ucmean	$-\infty$	-5.804	-6.761	-10.352
ucsv	$-\infty$	-6.761	-8.879	-7.851

3. Forecast performance — quarterly growth

The same scoring on each target's **single-quarter outcome** at t+h: the quarterly growth rate for real GDP and CPI inflation, and — since they are modelled in levels — the rate level for unemployment and the cash rate (identical to §2 for those two). Quarterly growth is a noisier, more local target than the accumulated level: it does not compound the path, so errors do not grow mechanically with horizon and the rankings can differ from the level view. Same conventions as §2.

Best single model by variable and horizon bucket (CRPS) — lowest mean CRPS in the bucket; value in parentheses:

Variable	near (1-4)	medium (5-8)	far (9-12)
gdp_growth	small_sv (0.679)	small_tight (0.773)	small_tight (0.846)
cpi_inflation	rw (0.189)	ar4 (0.273)	ucmean (0.302)

Variable	near (1-4)	medium (5-8)	far (9-12)
unemp_rate	medium_minn (0.325)	small_ss (0.631)	small_ss (0.726)
cash_rate	small_sv (0.307)	combo_equal (0.894)	ar4 (1.447)

Density accuracy by variable and horizon (CRPS, lower better)

Real GDP growth (qtr %) ([gdp_growth](#)) — quarterly growth rate (%)

Model	h=1	h=4	h=8	h=12
combo_equal	0.631	0.727	0.810	0.886
small_loose_p5	0.628	0.737	0.810	0.887
small_sv	0.604	0.726	0.840	0.908
ar4	0.668	0.727	0.797	0.898
small_ss	0.662	0.739	0.806	0.887
small_tight	0.682	0.743	0.794	0.878
medium_conj	0.672	0.757	0.800	0.876
small_minn	0.677	0.745	0.803	0.883
medium_minn	0.614	0.734	0.841	0.920
ucmean	0.680	0.732	0.816	0.908
ucsv	0.629	0.740	0.886	0.959
combo_pool	0.635	0.757	0.902	1.015
combo_logscore	0.760	0.855	1.132	1.043
combo_bma	0.840	0.931	1.411	1.165
rw	0.941	1.164	1.387	1.447

Trimmed-mean CPI inflation (qtr %) ([cpi_inflation](#)) — quarterly growth rate (%)

Model	h=1	h=4	h=8	h=12
ar4	0.162	0.246	0.288	0.314
ucmean	0.230	0.250	0.286	0.312
combo_equal	0.160	0.245	0.323	0.364
combo_pool	0.147	0.258	0.361	0.393
small_ss	0.172	0.268	0.354	0.369
rw	0.131	0.248	0.361	0.431
small_minn	0.170	0.262	0.355	0.389
combo_logscore	0.139	0.266	0.368	0.409
small_sv	0.171	0.270	0.359	0.406
small_tight	0.198	0.272	0.350	0.395
medium_minn	0.165	0.265	0.372	0.445
medium_conj	0.191	0.272	0.367	0.426
ucsv	0.176	0.285	0.386	0.430
combo_bma	0.146	0.291	0.407	0.437
small_loose_p5	0.193	0.296	0.414	0.420

Unemployment rate (%) (*unemp_rate*) – rate level at t+h (as \$2)

Model	h=1	h=4	h=8	h=12
small_ss	0.148	0.497	0.690	0.747
combo_equal	0.159	0.493	0.702	0.785
medium_minn	0.150	0.477	0.726	0.803
small_sv	0.157	0.501	0.743	0.771
small_minn	0.150	0.524	0.744	0.789
small_loose_p5	0.150	0.540	0.775	0.812
small_tight	0.158	0.549	0.779	0.811
combo_pool	0.177	0.496	0.767	0.860
ucsv	0.272	0.536	0.740	0.841
rw	0.169	0.545	0.777	0.913
medium_conj	0.163	0.545	0.820	0.896
ar4	0.189	0.534	0.760	0.955
combo_logscore	0.383	0.618	0.776	0.841
combo_bma	0.677	0.806	0.938	0.915
ucmean	0.726	0.774	0.905	1.034

Cash rate (%) (*cash_rate*) – rate level at t+h (as \$2)

Model	h=1	h=4	h=8	h=12
combo_equal	0.148	0.514	1.120	1.652
small_ss	0.140	0.526	1.149	1.673
ar4	0.167	0.589	1.141	1.593
rw	0.183	0.590	1.136	1.670
small_minn	0.144	0.528	1.152	1.792
combo_pool	0.129	0.549	1.225	1.715
combo_logscore	0.125	0.538	1.241	1.780
combo_bma	0.128	0.576	1.381	1.854
ucsv	0.175	0.646	1.281	1.842
small_tight	0.157	0.598	1.286	1.935
small_loose_p5	0.162	0.593	1.247	2.038
medium_conj	0.192	0.572	1.239	2.053
small_sv	0.113	0.535	1.410	2.147
medium_minn	0.112	0.559	1.465	2.494
ucmean	1.563	1.654	1.671	1.615

Point accuracy by variable and horizon (RMSE, lower better)

gdp_growth

Model	h=1	h=4	h=8	h=12
small_sv	1.573	1.664	1.806	1.912

Model	h=1	h=4	h=8	h=12
small_loose_p5	1.571	1.698	1.795	1.925
small_tight	1.614	1.687	1.784	1.907
small_ss	1.603	1.681	1.799	1.919
ucsv	1.618	1.675	1.812	1.909
small_minn	1.619	1.685	1.793	1.918
medium_conj	1.616	1.704	1.794	1.904
ar4	1.632	1.674	1.784	1.934
ucmean	1.619	1.684	1.801	1.929
combo_equal	1.624	1.681	1.816	1.921
medium_minn	1.605	1.676	1.819	1.946
combo_pool	1.688	1.721	1.895	2.011
combo_logscore	2.059	1.956	2.257	2.011
combo_bma	2.308	2.159	2.570	2.085
rw	2.377	2.326	2.609	2.661

cpi_inflation

Model	h=1	h=4	h=8	h=12
ar4	0.315	0.441	0.478	0.504
ucmean	0.434	0.449	0.480	0.507
combo_equal	0.327	0.471	0.581	0.603
combo_pool	0.287	0.482	0.614	0.628
small_ss	0.333	0.500	0.593	0.586
small_minn	0.333	0.498	0.594	0.610
combo_logscore	0.269	0.490	0.625	0.654
small_tight	0.402	0.508	0.586	0.612
rw	0.235	0.481	0.666	0.726
small_sv	0.360	0.514	0.632	0.654
combo_bma	0.278	0.527	0.687	0.684
ucsv	0.361	0.516	0.642	0.663
medium_conj	0.400	0.518	0.623	0.655
medium_minn	0.340	0.504	0.661	0.725
small_loose_p5	0.371	0.543	0.684	0.667

unemp_rate

Model	h=1	h=4	h=8	h=12
small_ss	0.339	0.890	1.208	1.226
ar4	0.413	0.915	1.174	1.300
combo_equal	0.370	0.903	1.259	1.324
small_sv	0.354	0.927	1.333	1.349

Model	h=1	h=4	h=8	h=12
small_minn	0.348	0.946	1.337	1.364
combo_pool	0.428	0.919	1.260	1.401
small_tight	0.362	0.953	1.351	1.371
small_loose_p5	0.349	0.989	1.382	1.467
medium_minn	0.341	0.943	1.388	1.516
ucsv	0.614	0.991	1.281	1.342
rw	0.391	0.984	1.391	1.509
medium_conj	0.375	1.050	1.487	1.574
combo_logscore	0.880	1.126	1.304	1.390
ucmean	1.260	1.326	1.435	1.542
combo_bma	1.252	1.365	1.502	1.448

cash_rate

Model	h=1	h=4	h=8	h=12
small_ss	0.276	0.990	1.908	2.581
ar4	0.307	1.103	1.931	2.467
combo_equal	0.341	0.971	1.916	2.777
combo_logscore	0.272	0.990	2.133	2.862
small_minn	0.292	1.051	2.045	2.902
rw	0.352	1.194	2.062	2.684
combo_pool	0.290	1.025	2.149	2.849
ucsv	0.358	1.208	2.080	2.695
combo_bma	0.282	1.054	2.353	2.961
small_tight	0.313	1.190	2.230	3.078
small_sv	0.252	0.982	2.306	3.468
small_loose_p5	0.352	1.113	2.180	3.387
medium_conj	0.479	1.218	2.280	3.418
medium_minn	0.265	1.052	2.486	4.079
ucmean	2.576	2.658	2.649	2.635

Density calibration — mean log predictive density (higher better)

Averaged across the 4 targets. $-\infty$ marks at least one origin where the realization fell outside a member's predictive support; the combinations, which always assign positive density, never do.

Model	h=1	h=4	h=8	h=12
combo_equal	-1.844	-1.514	-1.712	-1.878
combo_pool	-1.816	-1.654	-1.920	-2.039
combo_logscore	-1.925	-1.690	-1.982	-2.056
combo_bma	-3.731	-2.403	-2.354	-2.211
small_loose_p5	-6.452	-9.541	-9.016	-10.606

Model	h=1	h=4	h=8	h=12
ar4	$-\infty$	-7.791	-10.428	$-\infty$
medium_conj	$-\infty$	$-\infty$	$-\infty$	-13.455
medium_minn	$-\infty$	$-\infty$	-2.210	-5.106
rw	$-\infty$	-2.978	-2.956	-2.692
small_minn	$-\infty$	$-\infty$	$-\infty$	$-\infty$
small_ss	$-\infty$	-10.085	-12.188	-10.290
small_sv	-4.619	$-\infty$	-7.262	-3.452
small_tight	$-\infty$	-10.973	$-\infty$	-12.616
ucmean	$-\infty$	$-\infty$	$-\infty$	$-\infty$
ucsv	$-\infty$	$-\infty$	$-\infty$	-13.565

4. Do the combinations beat the best single model?

Mean CRPS over all 4 targets, by horizon bucket. The honest test of a pool is whether it beats both equal weights and the best individual member. (Buckets average errors of differing scale across horizons, so use this within a bucket to rank models, not to compare buckets.)

4a. Levels

Model	near (1-4)	medium (5-8)	far (9-12)
combo_equal	0.534	1.189	1.808
ar4	0.578	1.259	1.849
small_ss	0.554	1.276	1.910
small_sv	0.534	1.259	1.982
small_minn	0.562	1.286	1.946
medium_minn	0.525	1.266	2.100
small_tight	0.590	1.327	1.994
ucsv	0.594	1.312	2.036
medium_conj	0.588	1.323	2.052
small_loose_p5	0.580	1.356	2.082
combo_pool	0.549	1.333	2.184
combo_logscore	0.646	1.680	2.257
ucmean	1.027	1.576	2.042
combo_bma	0.780	2.199	2.662
rw	0.812	2.063	3.629

4b. Quarterly growth

Model	near (1-4)	medium (5-8)	far (9-12)
combo_equal	0.389	0.650	0.860
small_ss	0.396	0.665	0.862
ar4	0.414	0.668	0.870
small_minn	0.401	0.676	0.894
combo_pool	0.398	0.714	0.932

Model	near (1-4)	medium (5-8)	far (9-12)
small_tight	0.421	0.707	0.936
small_loose_p5	0.415	0.716	0.959
small_sv	0.388	0.715	0.988
medium_conj	0.423	0.706	0.968
ucsv	0.439	0.728	0.950
medium_minn	0.386	0.721	1.046
combo_logscore	0.461	0.768	0.945
rw	0.510	0.802	1.042
combo_bma	0.553	0.908	1.021
ucmean	0.831	0.896	0.950

5. Statistical significance (Diebold-Mariano)

How often each combination **significantly beats** the random-walk and AR(4) benchmarks on CRPS (Harvey-corrected, 10% level), counted over the 4 targets x 3 horizons {4, 8, 12} tested. A negative DM statistic means the combination is more accurate; significance is one-sided here. The tests have low power on these high-variance losses (dominated by a few episodes such as 2020), so the combinations beat the benchmarks on *average* (§4) more often than they do *significantly*.

5a. Levels

Combination	beats ar4	beats rw
combo_bma	0 / 12	1 / 12
combo_equal	2 / 12	1 / 12
combo_logscore	1 / 12	1 / 12
combo_pool	0 / 12	1 / 12

Strongest results (significant at 5% or better):

- `combo_equal` beats `ar4` on **unemp_rate** at h=12 (DM -3.66***)
- `combo_equal` beats `ar4` on **cash_rate** at h=4 (DM -2.22**)
- `combo_bma` beats `rw` on **gdp_growth** at h=4 (DM -2.19**)
- `combo_logscore` beats `ar4` on **unemp_rate** at h=12 (DM -2.14**)

5b. Quarterly growth

Combination	beats ar4	beats rw
combo_bma	0 / 12	1 / 12
combo_equal	2 / 12	1 / 12
combo_logscore	1 / 12	1 / 12
combo_pool	0 / 12	0 / 12

Strongest results (significant at 5% or better):

- `combo_equal` beats `ar4` on **unemp_rate** at h=12 (DM -3.66***)
- `combo_equal` beats `ar4` on **cash_rate** at h=4 (DM -2.22**)
- `combo_logscore` beats `ar4` on **unemp_rate** at h=12 (DM -2.14**)
- `combo_logscore` beats `rw` on **gdp_growth** at h=4 (DM -2.04**)

6. Model profiles

One entry per model: its specification, what makes it distinct, the role it plays in the suite, its strengths and failure modes, and where it actually ranks in this evaluation (the eval line is computed, not asserted). Full rationale is in README.md.

6a. VAR members

small_minn

Spec: Independent Normal-inverse-Wishart (Gibbs); 8-variable small SOE core; 4 lags; GLP marginal-likelihood shrinkage; constant volatility; LP scaling (COVID); sum-of-coefficients + dummy-initial-observation priors.

Distinctive: The workhorse Minnesota BVAR, estimated by Gibbs with an *independent* Normal-inverse-Wishart prior — the engine required to impose block exogeneity, since asymmetric (equation-specific) prior variances cannot be represented by a Kronecker/conjugate prior. Shrinkage is data-driven (GLP marginal-likelihood), and sum-of-coefficients + dummy-initial-observation priors discipline the I(1) levels (rates, real TWI).

Role: The central, representative small-SOE BVAR — the reference point the other small members are deliberate variations around (tighter, looser, steady-state, SV).

Strengths & failure modes: Solid all-rounder at short-to-medium horizons. Constant volatility means it leans on the LP scaling for 2020; without it the 2020 outliers would distort the coefficients.

In this evaluation — levels: strongest at cash_rate (medium (5-8)), ranked 2 of 11 individual models.

In this evaluation — quarterly growth: strongest at cash_rate (medium (5-8)), ranked 2 of 11 individual models.

See: README.md D3, D4, D5, D17

small_ss

Spec: Steady-state (Villani); 8-variable small SOE core; 4 lags; GLP marginal-likelihood shrinkage; constant volatility; LP scaling (COVID).

Distinctive: Reparametrised around its *unconditional means* (Villani steady state), with informative priors placed directly on long-run levels — inflation 2.5% (target midpoint), NAIRU 4.5%, neutral cash rate 3.5%, US potential growth — and on the foreign block's steady states, which the domestic forecast inherits.

Role: The **long-horizon anchor**. An iterated VAR reverts to its unconditional mean at $h = 8-12$; this member makes that mean economically grounded rather than the raw sample average.

Strengths & failure modes: Strongest at medium/far horizons for the mean-reverting targets. Vulnerable if a steady-state anchor is stale (e.g. a shifted neutral rate drags the long end); constant volatility under-disperses around 2020 absent the LP correction.

In this evaluation — levels: best individual model for unemp_rate at medium (5-8); unemp_rate at far (9-12); cash_rate at medium (5-8).

In this evaluation — quarterly growth: best individual model for unemp_rate at medium (5-8); unemp_rate at far (9-12); cash_rate at medium (5-8).

See: README.md D3, D4, D5, D17

small_sv

Spec: Stochastic volatility, equation-by-equation; 8-variable small SOE core; 4 lags; GLP marginal-likelihood shrinkage; stochastic volatility; t-errors (SV-t) (COVID).

Distinctive: Stochastic volatility estimated equation-by-equation (the Carriero-Clark-Marcellino triangular factorisation), with t-distributed errors (the CCMM SV-t COVID treatment). Block exogeneity is *exact* — foreign equations simply drop the domestic regressors.

Role: The **density-calibration specialist**: time-varying volatility tracks changing macro uncertainty and the fat tails absorb outliers instead of letting them widen the whole history.

Strengths & failure modes: Its SV captures the time-varying conditional variance, so it stays well-calibrated when uncertainty shifts — competitive on the GDP level at medium horizons. The volatility state at the jump-off can over/under-shoot if the last few quarters were unusual; the small system limits cross-variable information.

In this evaluation — levels: best individual model for gdp_growth at medium (5-8); gdp_growth at far (9-12); cash_rate at near (1-4).

In this evaluation — quarterly growth: best individual model for gdp_growth at near (1-4); cash_rate at near (1-4).

See: README.md D3, D6, D17

small_loose_p5

Spec: Independent Normal-inverse-Wishart (Gibbs); 8-variable small SOE core; 5 lags; fixed shrinkage $\lambda=0.4$; constant volatility; LP scaling (COVID).

Distinctive: A deliberately *under-shrunk*, longer-lag variant — fixed $\lambda = 0.4$ (vs the $\sim 0.1-0.15$ the GLP procedure selects) and 5 lags — so the data speak more and richer dynamics can show through, at the cost of estimation noise.

Role: The **loose / long-lag** diversity axis: it fails differently from the tightly-shrunk members and can capture dynamics they shrink away.

Strengths & failure modes: Occasionally best at near-horizon density when the extra flexibility pays; noisier and prone to wider intervals at long horizons (the cost of light shrinkage in a short sample).

In this evaluation — levels: strongest at gdp_growth (near (1-4)), ranked 4 of 11 individual models.

In this evaluation — quarterly growth: strongest at gdp_growth (near (1-4)), ranked 3 of 11 individual models.

See: README.md D4, D8

medium_minn

Spec: Stochastic volatility, equation-by-equation; 13-variable medium system; 2 lags; GLP marginal-likelihood shrinkage; stochastic volatility; t-errors (SV-t) (COVID).

Distinctive: The larger system — 13 variables (adds terms of trade, wages, employment, consumption, the 10y yield) with stochastic volatility and t-errors, but only 2 lags to keep the parameter count feasible; equation-by-equation estimation keeps the recursive loop tractable.

Role: The **medium-system** axis (Banbura-Giannone-Reichlin): medium systems often forecast best given enough shrinkage, and the extra variables bring cross-sectional information the small core lacks.

Strengths & failure modes: Strong short-horizon density (it tends to win the near bucket). The short lag length limits long-horizon dynamics, and more parameters mean more estimation uncertainty at the far end.

In this evaluation — levels: best individual model for gdp_growth at near (1-4); unemp_rate at near (1-4).

In this evaluation — quarterly growth: best individual model for unemp_rate at near (1-4).

See: README.md D1, D6, D17

medium_conj

Spec: Block-recursive conjugate NIW; 13-variable medium system; 4 lags; fixed shrinkage $\lambda=0.1$; constant volatility; LP scaling (COVID); sum-of-coefficients + dummy-initial-observation priors.

Distinctive: The medium system estimated by the fast *block-recursive conjugate* scheme (foreign VAR + domestic block conditioned on contemporaneous foreign values, the RBNZ Bloor-Matheson approach) — closed-form, so cheap even at 13 variables x 4 lags. Block exogeneity is exact by the recursive structure, not the prior.

Role: The cheap medium workhorse; it complements `medium_minn` (conjugate constant-volatility vs SV) on the same large system.

Strengths & failure modes: Tends to lead the far-horizon GDP level error. Constant volatility leans on LP for 2020; the conjugate Kronecker prior cannot represent asymmetric shrinkage, which is why block exogeneity comes from the recursive structure.

In this evaluation — levels: strongest at gdp_growth (far (9-12)), ranked 4 of 11 individual models.

In this evaluation — quarterly growth: strongest at gdp_growth (far (9-12)), ranked 2 of 11 individual models.

See: README.md D3, D5, D8

small_tight

Spec: Block-recursive conjugate NIW; 8-variable small SOE core; 4 lags; fixed shrinkage $\lambda=0.05$; constant volatility; LP scaling (COVID); sum-of-coefficients + dummy-initial-observation priors.

Distinctive: The heavily-shrunk small model — fixed $\lambda = 0.05$, far tighter than the GLP selection — pulling hard toward the persistence/random-walk prior, so it is parsimonious and low-variance.

Role: The **tight** diversity axis and the long-horizon robustness member: heavy shrinkage buys stability where lightly-parametrised models wander.

Strengths & failure modes: Best or near-best at far-horizon GDP (the tight prior stops it over-reacting). Can be too rigid at short horizons, missing genuine dynamics the looser members catch.

In this evaluation — levels: strongest at gdp_growth (far (9-12)), ranked 2 of 11 individual models.

In this evaluation — quarterly growth: best individual model for gdp_growth at medium (5-8); gdp_growth at far (9-12).

See: README.md D4, D5, D8

6b. Benchmark members

rw

Spec: Random walk; LP scaling (COVID).

Distinctive: The no-change forecast: the last observed value persists, with Gaussian increments scaled to the historical change.

Role: The universal hard-to-beat short-horizon bar for persistent/level variables, and a pool member.

Strengths & failure modes: Competitive at $h = 1$ for level variables; fails badly at long horizons — its level path runs away, which the level-error tables (§2) expose brutally.

In this evaluation — levels: best individual model for cpi_inflation at near (1-4); cpi_inflation at medium (5-8).

In this evaluation — quarterly growth: best individual model for cpi_inflation at near (1-4).

See: README.md D8

ar4

Spec: Bayesian AR(4); LP scaling (COVID).

Distinctive: A Bayesian AR(4) per variable with Minnesota-style lag shrinkage and a stationarity-truncated posterior.

Role: The univariate-persistence bar — it isolates how much of the forecast is just own-history dynamics.

Strengths & failure modes: Surprisingly strong for inflation at near/medium horizons, where univariate dynamics dominate; it cannot use cross-variable information, so it lags when that matters.

In this evaluation — levels: best individual model for cpi_inflation at far (9-12); cash_rate at far (9-12).

In this evaluation — quarterly growth: best individual model for cpi_inflation at medium (5-8); cash_rate at far (9-12).

See: README.md D8

ucsv

Spec: Unobserved components + stochastic volatility; t-errors (robust) (COVID).

Distinctive: Stock-Watson unobserved-components stochastic volatility per variable: a random-walk trend plus transitory noise, both with time-varying variances and outlier-robust t-errors.

Role: The canonical inflation benchmark and a genuine density anchor for the other targets.

Strengths & failure modes: Strong for inflation (its native use case); weaker for variables with richer multivariate dynamics. The trend/noise split is weakly identified, so it is gated on Monte-Carlo precision, not raw ESS.

In this evaluation — levels: strongest at gdp_growth (near (1-4)), ranked 3 of 11 individual models.

In this evaluation — quarterly growth: strongest at gdp_growth (near (1-4)), ranked 4 of 11 individual models.

See: README.md D9, D17

ucmean

Spec: Unconditional mean; LP scaling (COVID).

Distinctive: A Gaussian density centred on the expanding-sample mean with the sample variance — the simplest possible density forecast.

Role: The floor: the ‘did the model beat just predicting the long-run average’ bar.

Strengths & failure modes: Unexpectedly competitive at long horizons for mean-reverting growth (everything reverts to the mean eventually); useless at short horizons where dynamics matter.

In this evaluation — levels: strongest at cpi_inflation (far (9-12)), ranked 3 of 11 individual models.

In this evaluation — quarterly growth: best individual model for cpi_inflation at far (9-12).

See: README.md D8

6c. Combination schemes

`combo_equal`

Weights: Equal weights on every member, per variable x horizon bucket.

In this evaluation: The forecast-combination-puzzle benchmark and the recommended robust default: in the evaluation it has the best mean log score at every horizon and the best far-horizon CRPS. Hard to beat because it never over-fits weights.

`combo_logscore`

Weights: Weights proportional to each member’s recent log predictive score, with a forgetting factor, shrunk toward equal.

In this evaluation: Adapts to which members are forecasting well lately; the shrinkage and forgetting guard against over-concentrating on a member that was lucky.

`combo_pool`

Weights: Optimal prediction pool (Hall-Mitchell / Geweke-Amisano): weights on the simplex that maximise the historical *pooled* log score, shrunk toward equal.

In this evaluation: Unlike BMA it does not degenerate to a single model (‘all models are false but useful’); competitive with equal weights and occasionally better at near horizons.

`combo_bma`

Weights: Bayesian model averaging by predictive likelihood — no shrinkage.

In this evaluation: **Diagnostic only.** It concentrates weight on the single best-fitting member, so it answers ‘which model does the data favour’ rather than serving as a robust combination; reported, not recommended.

7. How to read this

- **Two complementary metrics.** The **level** error (§2) is the headline: for GDP and inflation it is the cumulative level from the forecast origin (where real GDP and the price level land h quarters out), for unemployment and the cash rate the rate level at t+h. It accumulates the whole forecast path, so it grows with horizon and exposes a model that gets the persistent/drift component wrong (e.g. the random walk). The **quarterly-growth** view (§3) scores each target’s single-quarter outcome — a noisier, more local target that does not compound the path, so it does not grow mechanically with horizon and can rank models differently. For unemployment and the cash rate (modelled in levels) the two views coincide. The year-ended *growth* scores live in [output/tables/scores_by_horizon.csv](#).
- **Point gains over the best member are modest by design** — equal weights are hard to beat (the forecast-combination puzzle). The pool’s payoff is *calibration and robustness*: it insures against any single member failing, rather than always winning on accuracy.
- **CRPS and log score can disagree** on outlier-heavy windows (log score is far more sensitive to tail events); both are reported. A COVID-excluded variant is in [output/tables/scores_by_horizon_excovid.csv](#). See the full Quarto report for fan charts, PIT calibration, and weight-evolution plots.